Project Python Foundations: FoodHub Data Analysis

Marks: 60

Context

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. Online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company FoodHub offers access to multiple restaurants through a single smartphone app.

The app allows the restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirms the pick-up in the app and travels to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin of the delivery order from the restaurants.

Objective

The food aggregator company has stored the data of the different orders made by the registered customers in their online portal. They want to analyze the data to get a fair idea about the demand of different restaurants which will help them in enhancing their customer experience. Suppose you are hired as a Data Scientist in this company and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company to improve the business.

Data Description

The data contains the different data related to a food order. The detailed data dictionary is given below.

Data Dictionary

- order_id: Unique ID of the order
- customer_id: ID of the customer who ordered the food
- restaurant_name: Name of the restaurant
- cuisine_type: Cuisine ordered by the customer
- cost: Cost of the order
- day_of_the_week: Indicates whether the order is placed on a weekday or weekend (The weekday is from Monday to Friday and the weekend is Saturday and Sunday)
- rating: Rating given by the customer out of 5
- food_preparation_time: Time (in minutes) taken by the restaurant to prepare the food. This is calculated by taking the difference between the timestamps of the restaurant's order confirmation and the delivery person's pick-up confirmation.
- delivery_time: Time (in minutes) taken by the delivery person to deliver the food package. This is calculated by taking the difference between the timestamps of the delivery person's pick-up confirmation and drop-off information

Let us start by importing the required libraries

In [57]:
```
# Installing the libraries with the specified version.
!pip install numpy==2.0.2 pandas==2.2.2 matplotlib==3.10.0 seaborn==0.13.2 -q --user
```

In [58]:
```
# import libraries for data manipulation
import numpy as np
import pandas as pd

# import libraries for data visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

Understanding the structure of the data

In [59]:
```
 # uncomment and run the following lines for Google Colab
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

In [60]:
```
# read the data
df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/foodhub_order.csv')
# returns the first 5 rows
df.head()
```

Out[60]:

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24

Observations:

The DataFrame has 9 columns as mentioned in the Data Dictionary. Data in each row corresponds to the order placed by a customer.

Question 1: How many rows and columns are present in the data? [0.5 mark]

In [61]:
```
# find the shape of the data set
print("There are", df.shape[0], 'rows and', df.shape[1], "columns.")
```

```
There are 1898 rows and 9 columns.
```

Observations:

The data has 1898 rows and 9 columns

Question 2: What are the datatypes of the different columns in the dataset? (The info() function can be used) [0.5 mark]

In [62]:
```
# Use info() to print a concise summary of the DataFrame
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   order_id               1898 non-null   int64
 1   customer_id            1898 non-null   int64
 2   restaurant_name        1898 non-null   object
 3   cuisine_type           1898 non-null   object
 4   cost_of_the_order      1898 non-null   float64
 5   day_of_the_week        1898 non-null   object
 6   rating                 1898 non-null   object
 7   food_preparation_time  1898 non-null   int64
 8   delivery_time          1898 non-null   int64
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

Observations:

- There are 1898 non-null values in the each column
- The data contains integer, float, and object type data
  - Integer: order_id, customer_id, food_preparation_time, and delivery_time
  - Float: cost_of_the_order
  - object: restaurant_name, cuisine_type, day_of_the_week, rating
- There's a total of 9 columns
- Total memort is 133.6+ KB

Question 3: Are there any missing values in the data? If yes, treat them using an appropriate method. [1 mark]

In [63]:
```
# using isnull() function to find missing values
df.isnull().sum()
```

Out[63]:

	0
order_id	0
customer_id	0
restaurant_name	0
cuisine_type	0
cost_of_the_order	0
day_of_the_week	0
rating	0
food_preparation_time	0
delivery_time	0

dtype: int64

Observations:

There are no missing values in any of the columns of the data

Question 4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed? [2 marks]

In [64]:
```
# get summary statistics
df.describe().T
```

Out[64]:

	count	mean	std	min	25%	50%	75%	max
order_id	1898.0	1.477496e+06	548.049724	1476547.00	1477021.25	1477495.50	1.477970e+06	1478444.00
customer_id	1898.0	1.711685e+05	113698.139743	1311.00	77787.75	128600.00	2.705250e+05	405334.00
cost_of_the_order	1898.0	1.649885e+01	7.483812	4.47	12.08	14.14	2.229750e+01	35.41
food_preparation_time	1898.0	2.737197e+01	4.632481	20.00	23.00	27.00	3.100000e+01	35.00
delivery_time	1898.0	2.416175e+01	4.972637	15.00	20.00	25.00	2.800000e+01	33.00

Observations:

- The minimym value for food prepartion is 20 minutes
- The average time it takes for food to be prepared is 27.37 minutes
- the maximum time it takes for food to be prepared is 35 mins

Question 5: How many orders are not rated? [1 mark]

In [65]:
```
# pulling up each unique value using value counts, then observe the number of not given ratings
df['rating'].value_counts()
```

Out[65]:

	count
rating	
Not given	736
5	588
4	386
3	188

dtype: int64

Observations:

- There are 736 orders that are not rated

Exploratory Data Analysis (EDA)

Univariate Analysis

Question 6: Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, countplots, etc. are used for univariate exploration.) [9 marks]

In [66]:
```
df['order_id'].nunique()
```

Out[66]: `1898`

In [67]:
```
df['customer_id'].nunique()
```

Out[67]: `1200`

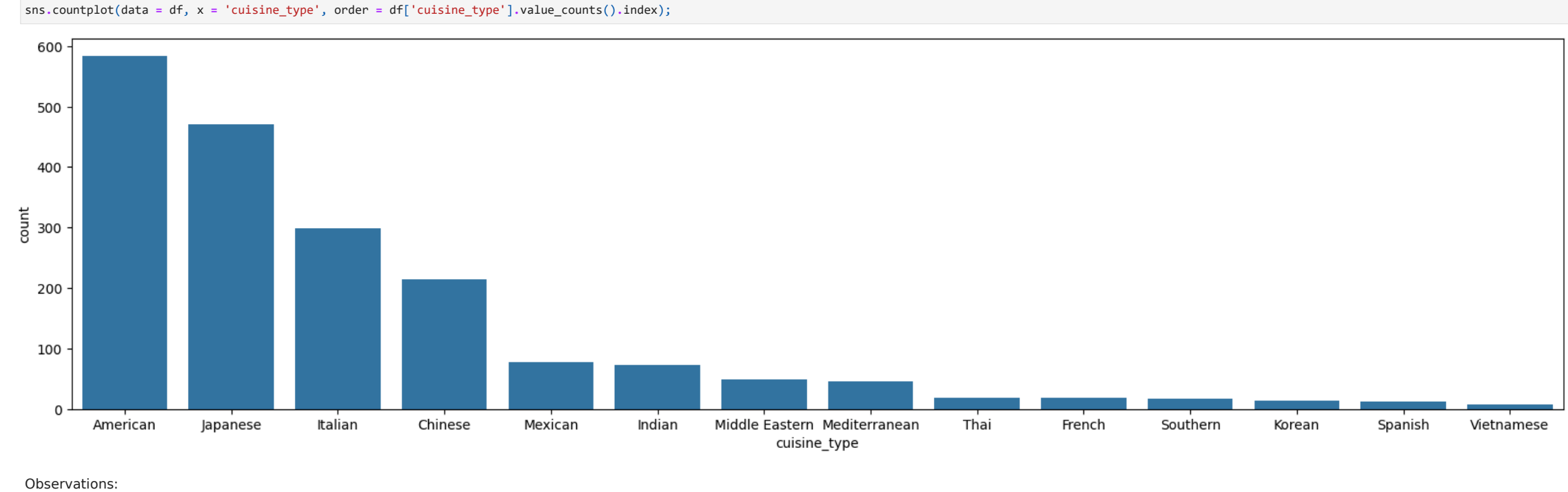
In [68]:
```
df['restaurant_name'].nunique()
```

Out[68]: `178`

In [69]:
```
df['cuisine_type'].nunique()
```

Out[69]: `14`

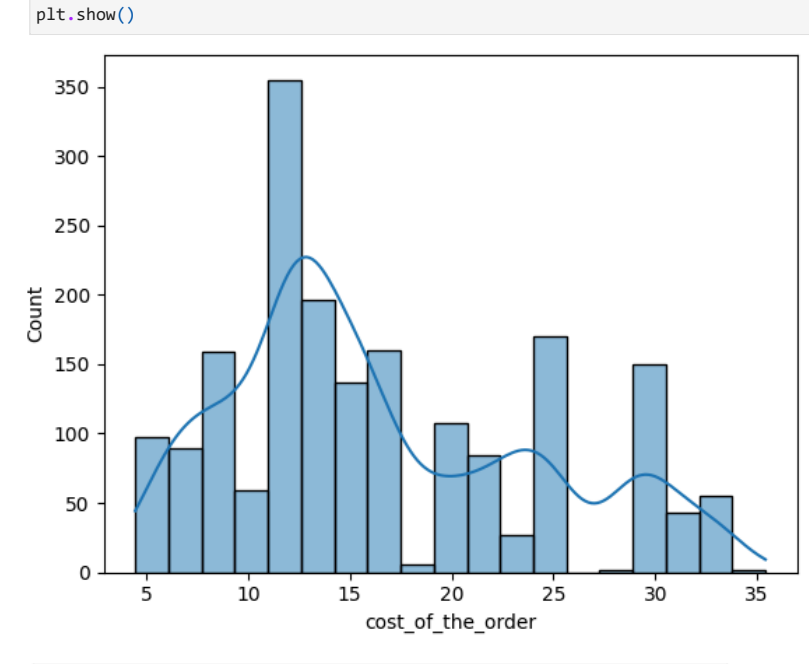
In [70]:
```
# bar graph of cuisine type
plt.figure(figsize = (20,5))
sns.countplot(data = df, x = 'cuisine_type', order = df['cuisine_type'].value_counts().index);
```

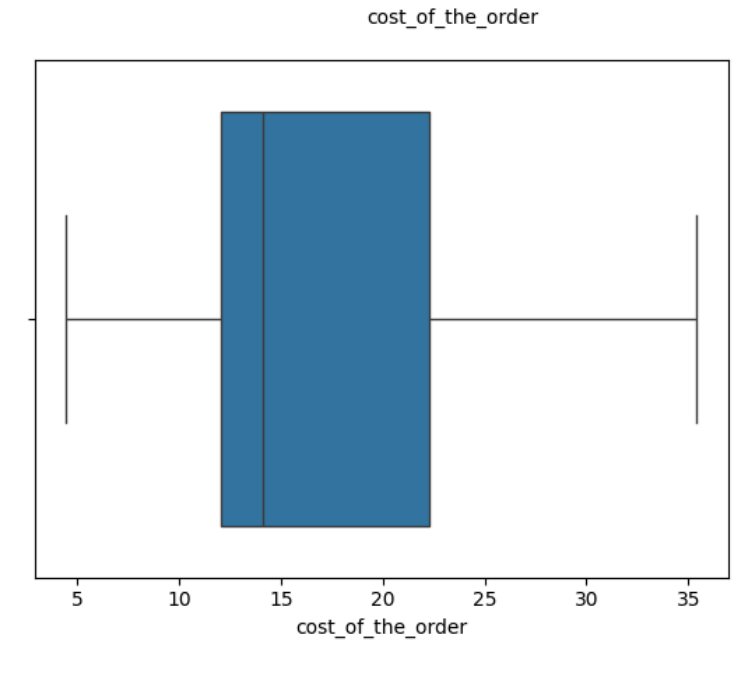


Observations:

- The data appears to be skewed right
- American, Japanese, and Italian are the top 3 cuisines and have a large gap on the rest of the restuarants
- Vietnamese is the last of cuisines

In [71]:
```
#histogram and boxplot of cost of the order
sns.histplot(data = df, x = 'cost_of_the_order', kde = True)
plt.show()
sns.boxplot(data = df, x = 'cost_of_the_order')
plt.show()
```





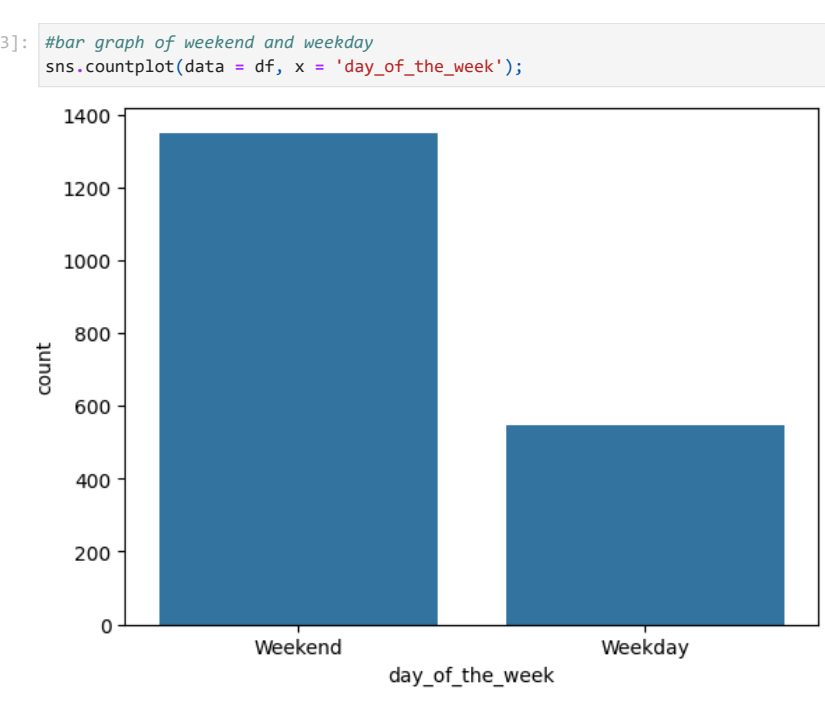
Observations:

- The data appears to be skewed right, given that the mean is greater than the median
- 12-13 dollars appears to be the most rated value
- Most of the orders appear to be cheap, since there are few orders that go above 30 dollars. This could mean that most customers perfer cheaper meals on Foodhub

In [72]:
```
df['day_of_the_week'].nunique()
```

Out[72]: `2`

In [73]:
```
#bar graph of weekend and weekday
sns.countplot(data = df, x = 'day_of_the_week');
```



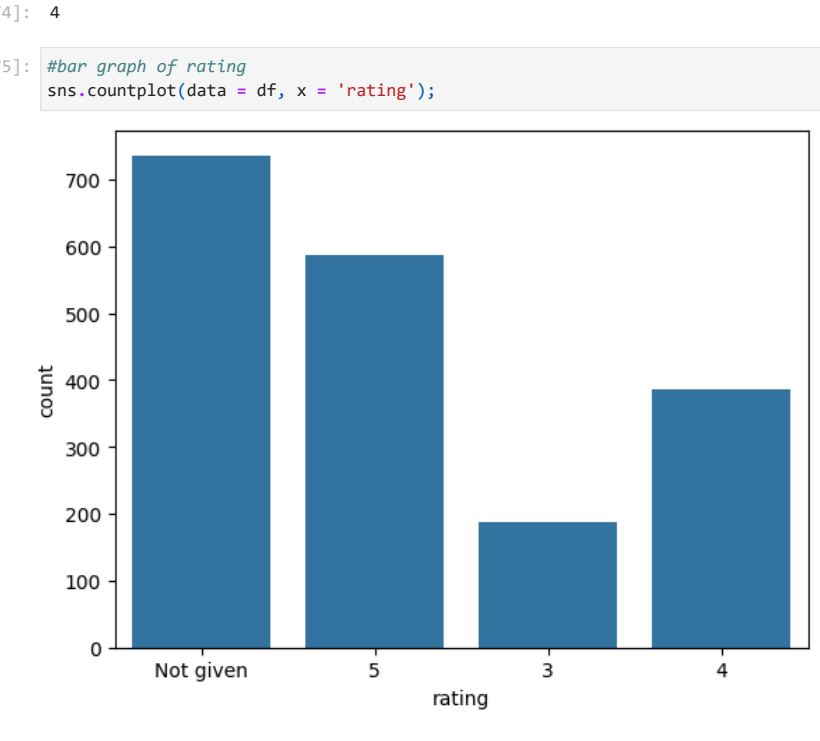
Observations:

- The weekend has a huge influx of orders
- The weekend brings in twice the amount of orders compared to the Weekday

In [74]:
```
df['rating'].nunique()
```

Out[74]: `4`

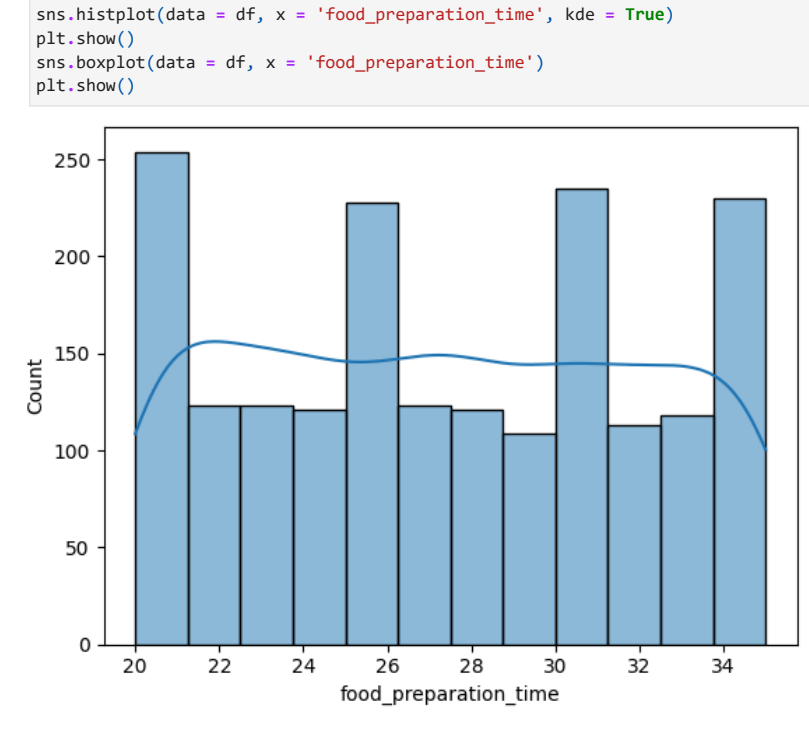
In [75]:
```
#bar graph of rating
sns.countplot(data = df, x = 'rating');
```

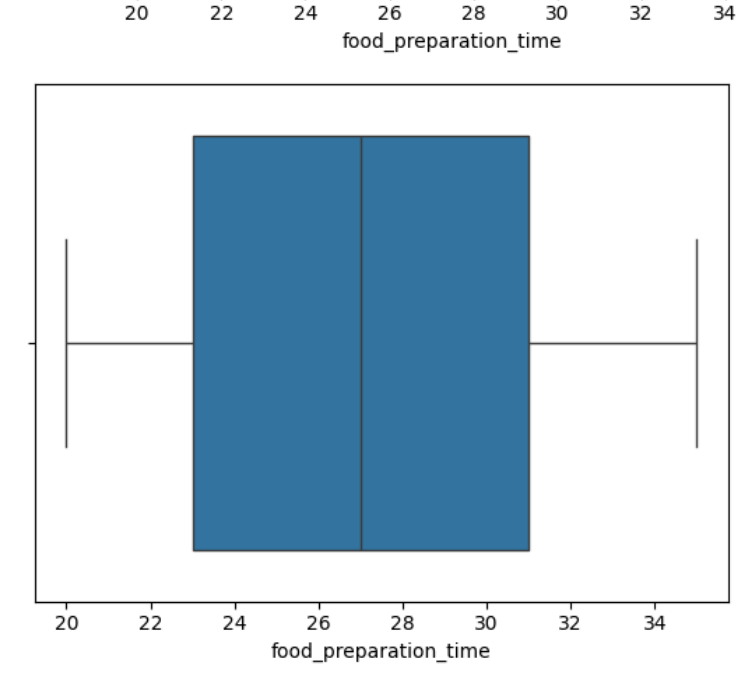


Observations:

- The majority of orders go unrated
- The lowest rating showed is a 3, meaning the almost all customers are satisfied with their orders

In [76]:
```
# histogram and boxplot of food preparation time
sns.histplot(data = df, x = 'food_preparation_time', kde = True)
plt.show()
sns.boxplot(data = df, x = 'food_preparation_time')
plt.show()
```

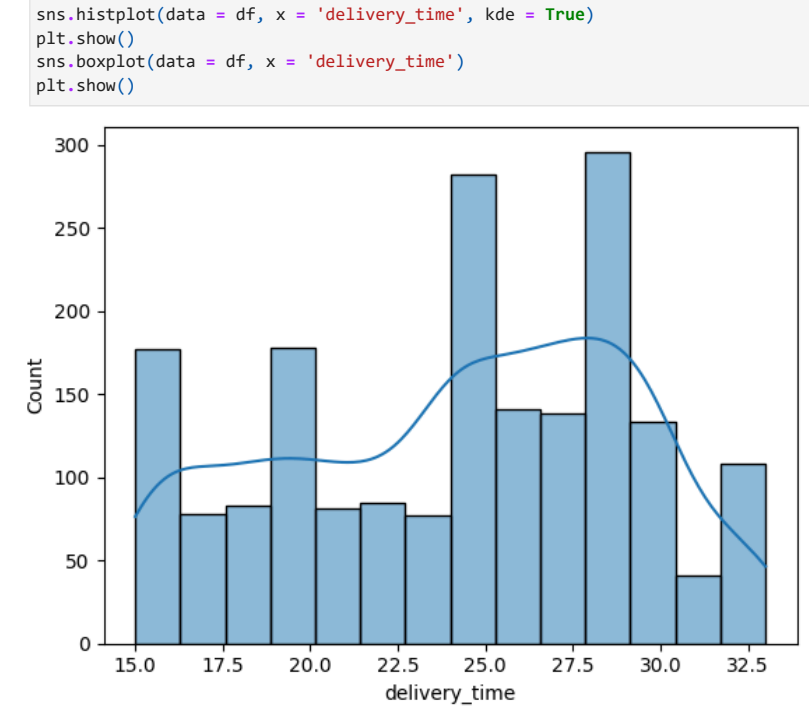


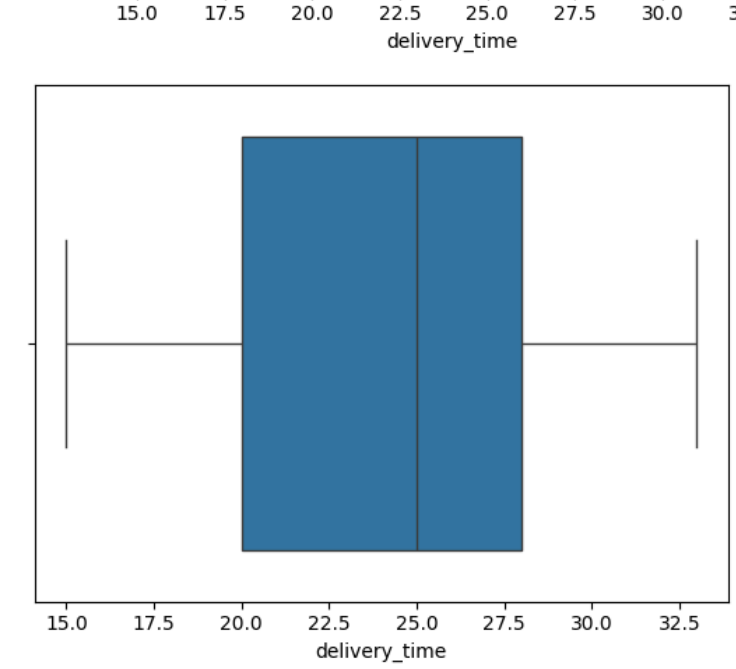


Observations:

- The graph display a normal distribution, given that the median mean are close to each other
- There are no outliers in this data set

In [77]:
```
# histogram and box plot of delivery time
sns.histplot(data = df, x = 'delivery_time', kde = True)
plt.show()
sns.boxplot(data = df, x = 'delivery_time')
plt.show()
```





Observations:

- The average delivery time appears to be slightly skewed left, since the mean is less than the median
- The majority of the orders appear to be around 25-30 minutes

Question 7: Which are the top 5 restaurants in terms of the number of orders received? [1 mark]

In [78]:
```
# obtain top 5 restaurants with the highest number of orders
df['order_count'] = 1
top5 = df.groupby(by = 'restaurant_name').sum()
top5_restaurants= df.groupby('restaurant_name').sum()[['order_count']].sort_values('order_count', ascending = False).reset_index().head(5)
top5_restaurants
```

Out[78]:

	restaurant_name	order_count
0	Shake Shack	219
1	The Meatball Shop	132
2	Blue Ribbon Sushi	119
3	Blue Ribbon Fried Chicken	96
4	Parm	68

Observations:

- The top 5 restaurants are Shake Shack, The Meatball Shop, Blue Ribbon Sushi, Blue Ribbon Fried Chicken, and Parm
- 1st rated restaurant and 5th rated restuarnat have values of 219 and 68 respectively, indicating a large range
- The top 5 restuarants take up roughly a third of all orders

Question 8: Which is the most popular cuisine on weekends? [1 mark]

In [79]:
```
#Find the most popular cuisine on weekend
df_weeknd = df[df['day_of_the_week'] == 'Weekend']
df_weeknd['cuisine_type'].value_counts()
```

Out[79]:

	count
cuisine_type	
American	415
Japanese	335
Italian	207
Chinese	163
Mexican	53
Indian	49
Middle Eastern	32
Mediterranean	32
Thai	15
French	13
Korean	11
Southern	11
Spanish	11
Vietnamese	4

dtype: int64

Observations:

- The most popular cuisine type on weekends is American
- The top 5 cuisines on weekends are American, Japanese, Italian, Chinese, and Mexican
- The top restauarnt had 415 orders on the weekend and the lowest restuarant had 4 orders, inidicating a wide range of values

Question 9: What percentage of the orders cost more than 20 dollars? [2 marks]

In [80]:
```
#Find the orders that cost more than $20
df_greater_than_20 = df[df['cost_of_the_order'] > 20]
#Divide the orders that cost more than $20 by the the total orders and multiply by 100
percentage = (df_greater_than_20.shape[0] / df.shape[0]) * 100
print('The percentage of the orders that cost more than $20 is', round(percentage, 2), '%')
```

```
The percentage of the orders that cost more than $20 is 29.24 %
```

Observations:

- The percenetage of orders that cost more than $20 is roughly 29%
- This indicates that a vast majority of the orders are under 20 dollars

Question 10: What is the mean order delivery time? [1 mark]

In [81]:
```
#Find the mean delivery time
mean_delivery_time = df['delivery_time'].mean()
print('The mean order delivery time is:', round(mean_delivery_time, 2), 'minutes')
```

```
The mean order delivery time is: 24.16 minutes
```

Observations:

- The mean delivery time is roughly 24 minutes
- This could be due to outside factors, such as distance for the driver to the customer, or traffic.

Question 11: The company has decided to give 20% discount vouchers to the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed. [1 mark]

In [82]:
```
#Find the top 5 most frequent customers
df['customer_id'].value_counts().head(5)
```

Out[82]:

	count
customer_id	
52832	13
47440	10
83287	9
250494	8
259341	7

dtype: int64

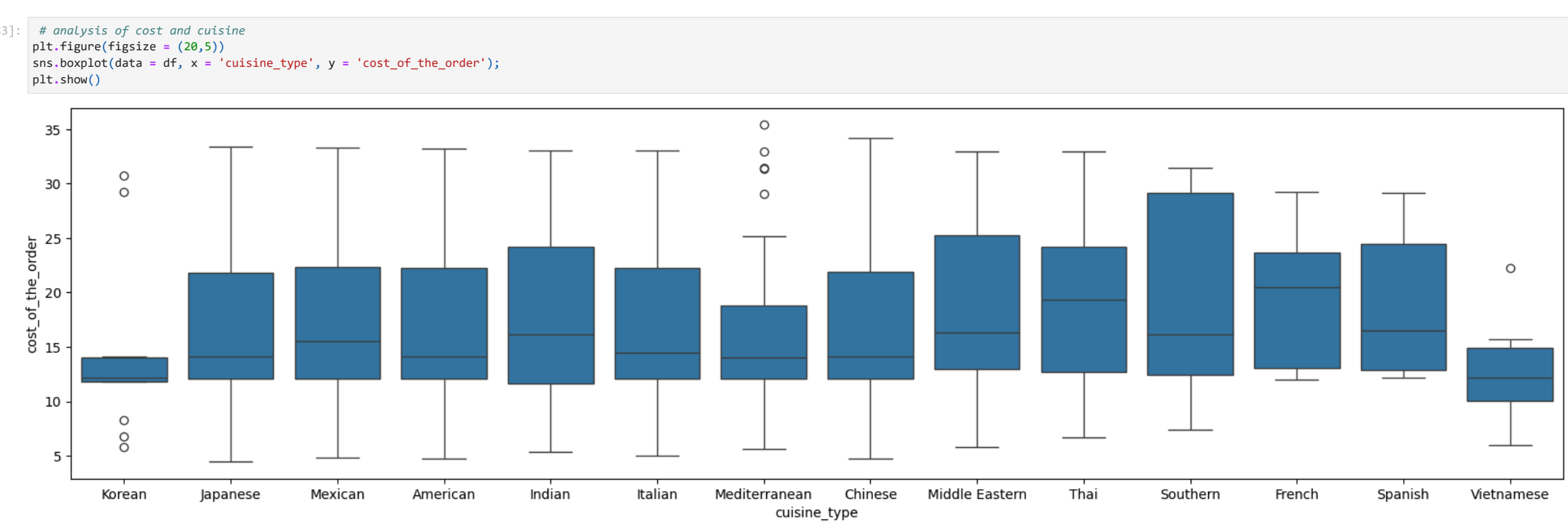
Observations:

- The amount of orders made by the top 5 most frequent customers adds up 47
- Customer 53832 has the most orders with 13
- Customer 259341 has the fifth most with 7

Multivariate Analysis

Question 12: Perform a multivariate analysis to explore relationships between the important variables in the dataset. (It is a good idea to explore relations between numerical variables as well as relations between numerical and categorical variables) [10 marks]

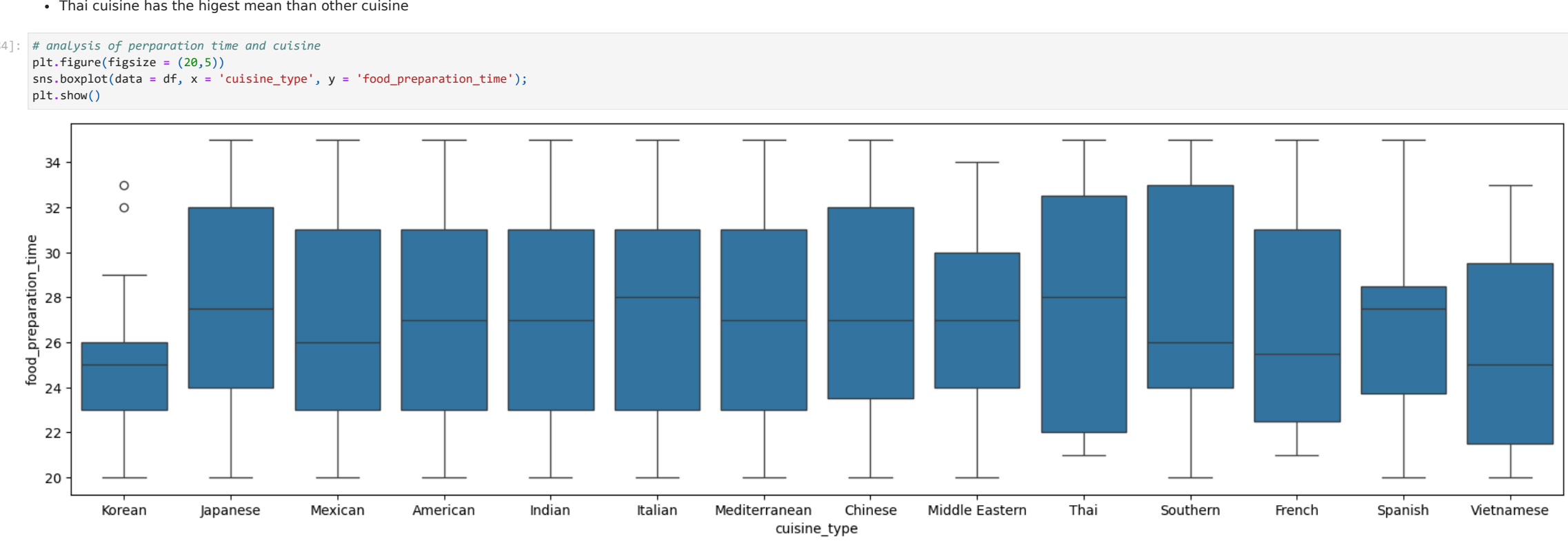
In [83]:
```
 # analysis of cost and cuisine
plt.figure(figsize = (20,5))
sns.boxplot(data = df, x = 'cuisine_type', y = 'cost_of_the_order');
plt.show()
```



Observations:

- There are muliple outliers in Korean, Mediterranean, and Vietnamese cuisines
- Korean and Vietnamese cuisines appears to be the cheapest cuisines
- Thai cuisine has the higest mean than other cuisine

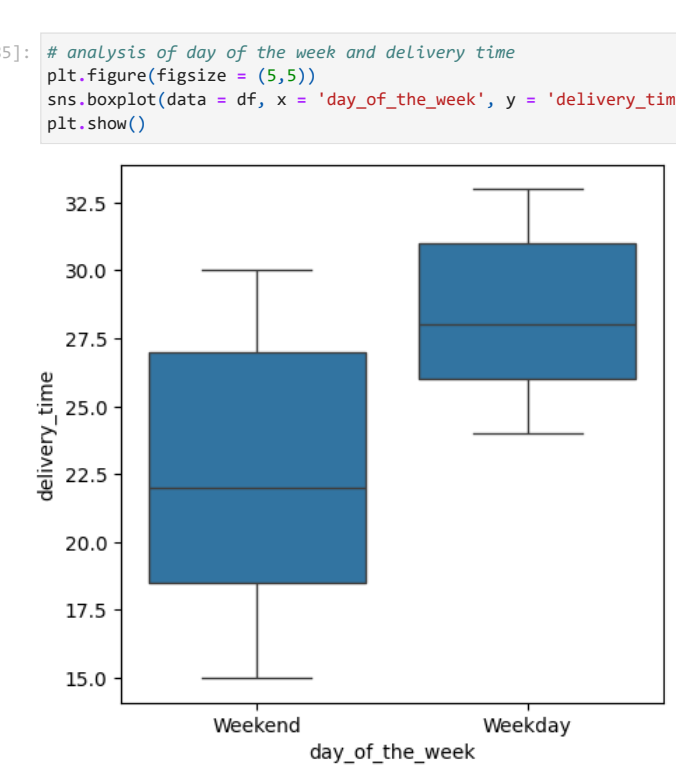
In [84]:
```
# analysis of perparation time and cuisine
plt.figure(figsize = (20,5))
sns.boxplot(data = df, x = 'cuisine_type', y = 'food_preparation_time');
plt.show()
```



Observations:

- Food prepartion appears to be similar in times across all cuisines
- Thai has the highest mean than any other cuisine
- Korean cuisine contains outliers

In [85]:
```
# analysis of day of the week and delivery time
plt.figure(figsize = (5,5))
sns.boxplot(data = df, x = 'day_of_the_week', y = 'delivery_time');
plt.show()
```



Observations:

- Weekday has a higher average delivery time than weekend
- Weekend overall appears to be lower in delivery time compared to weekday

In [86]:
```
df.groupby(['restaurant_name'])['cost_of_the_order'].mean().sort_values(ascending = False).reset_index()
```

Out[86]:

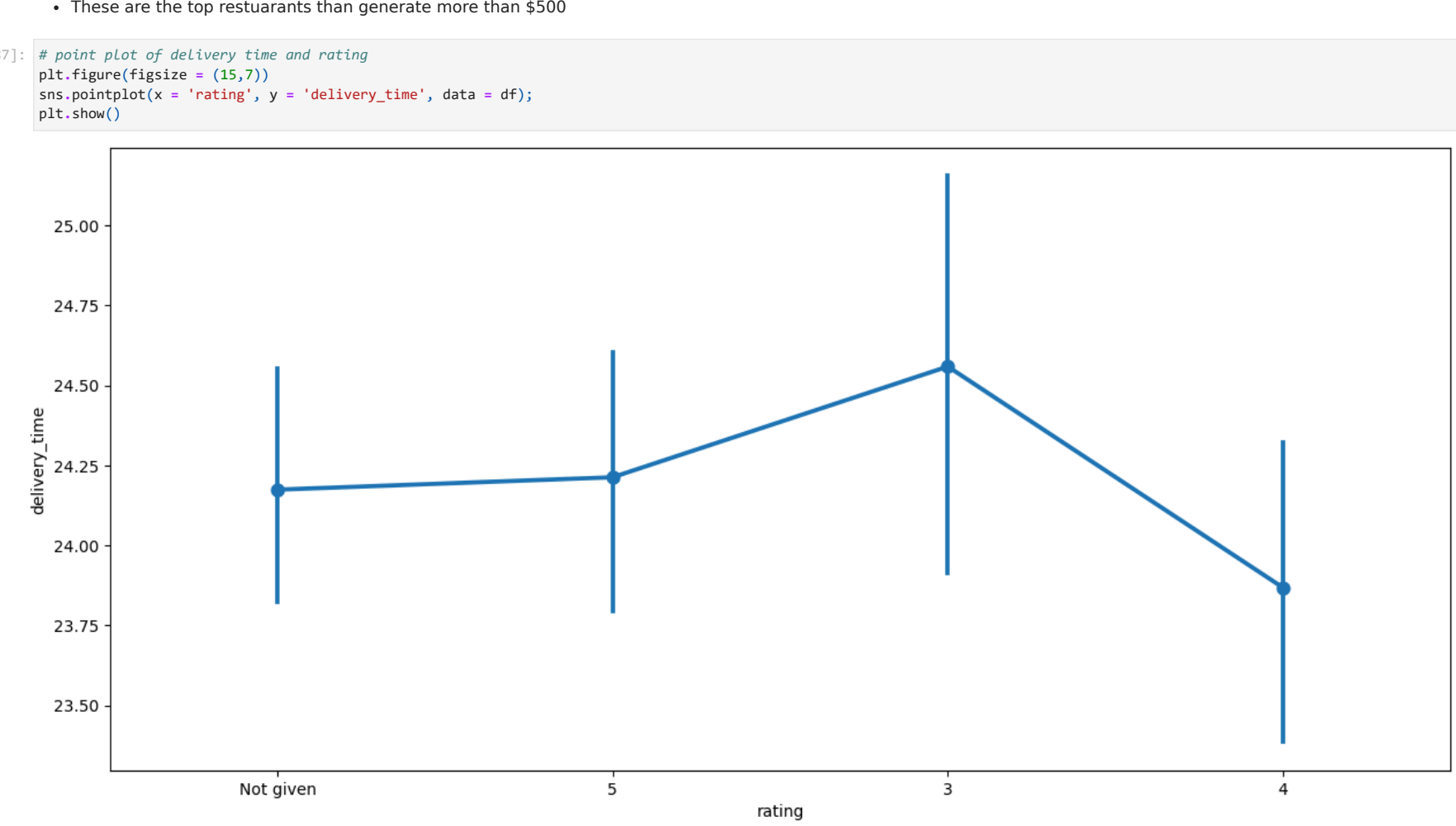
	restaurant_name	cost_of_the_order
0	Kambi Ramen House	32.930
1	Emporio	31.430
2	Bhatti Indian Grill	31.115
3	Haru Gramercy Park	29.830
4	Il Bambino	29.250
...	...	...
173	Market Table	6.790
174	Wa Jeal	6.740
175	Big Wong Restaurant ç_¤¾Ñ¼	5.920
176	Cipriani Le Specialita	5.920
177	Hunan Manor	5.720

178 rows × 2 columns

Observations

- These are the top restuarants than generate more than $500

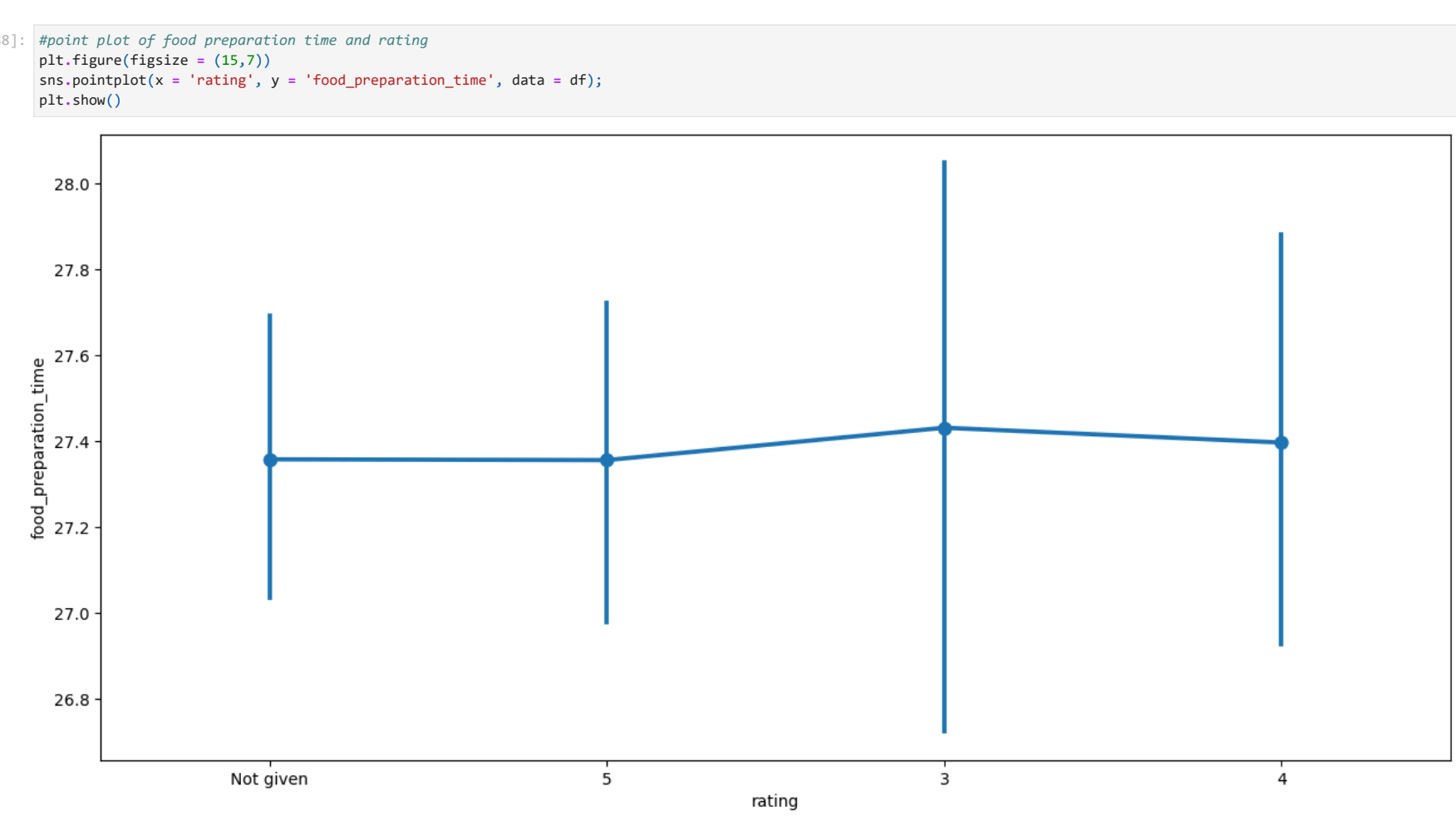
In [87]:
```
# point plot of delivery time and rating
plt.figure(figsize = (15,7))
sns.pointplot(x = 'rating', y = 'delivery_time', data = df);
plt.show()
```



Observations:

- There might be a rcoreelation between high delivery times and customer satisfaction

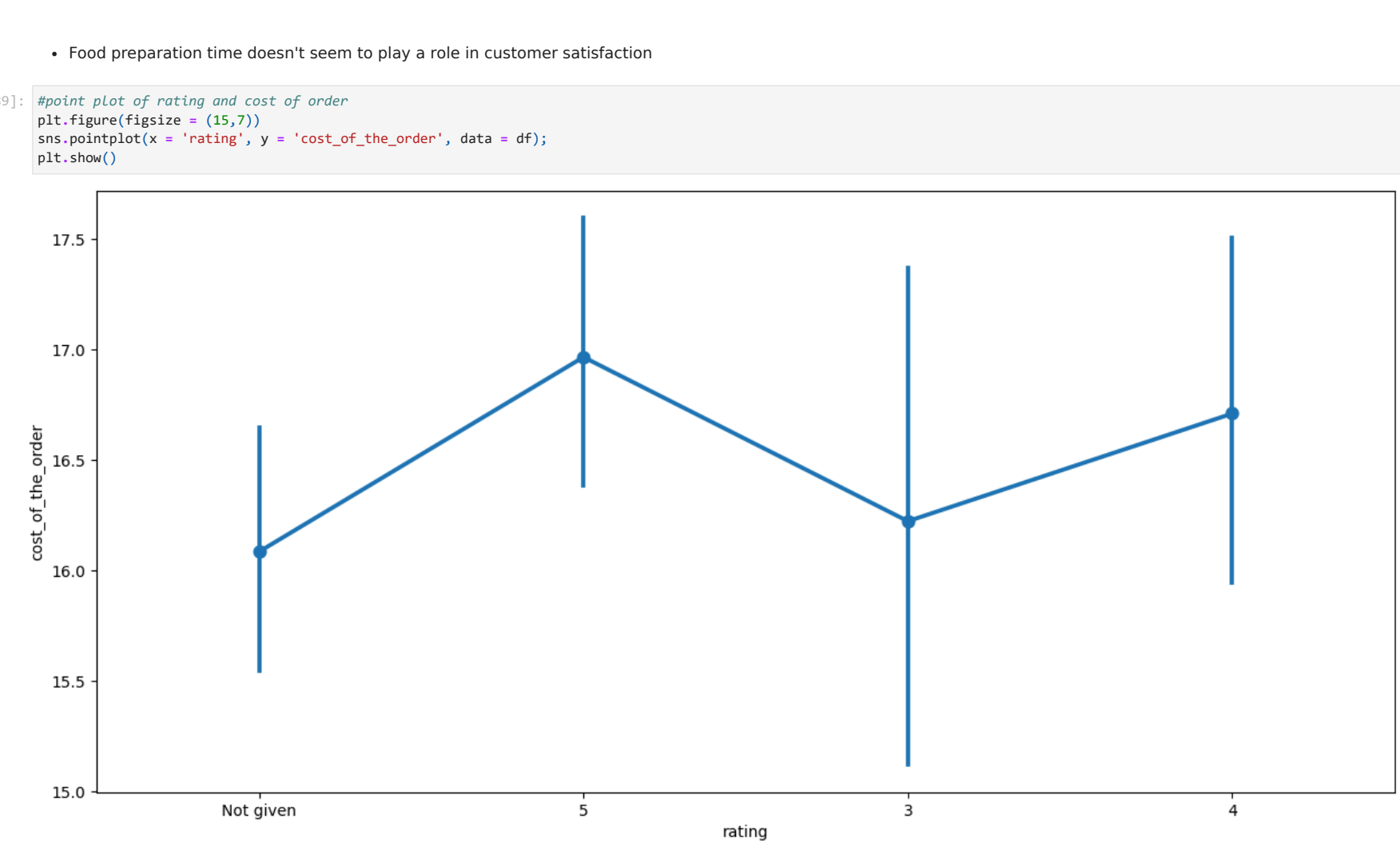
In [88]:
```
#point plot of food preparation time and rating
plt.figure(figsize = (15,7))
sns.pointplot(x = 'rating', y = 'food_preparation_time', data = df);
plt.show()
```



Observations:

- Food preparation time doesn't seem to play a role in customer satisfaction

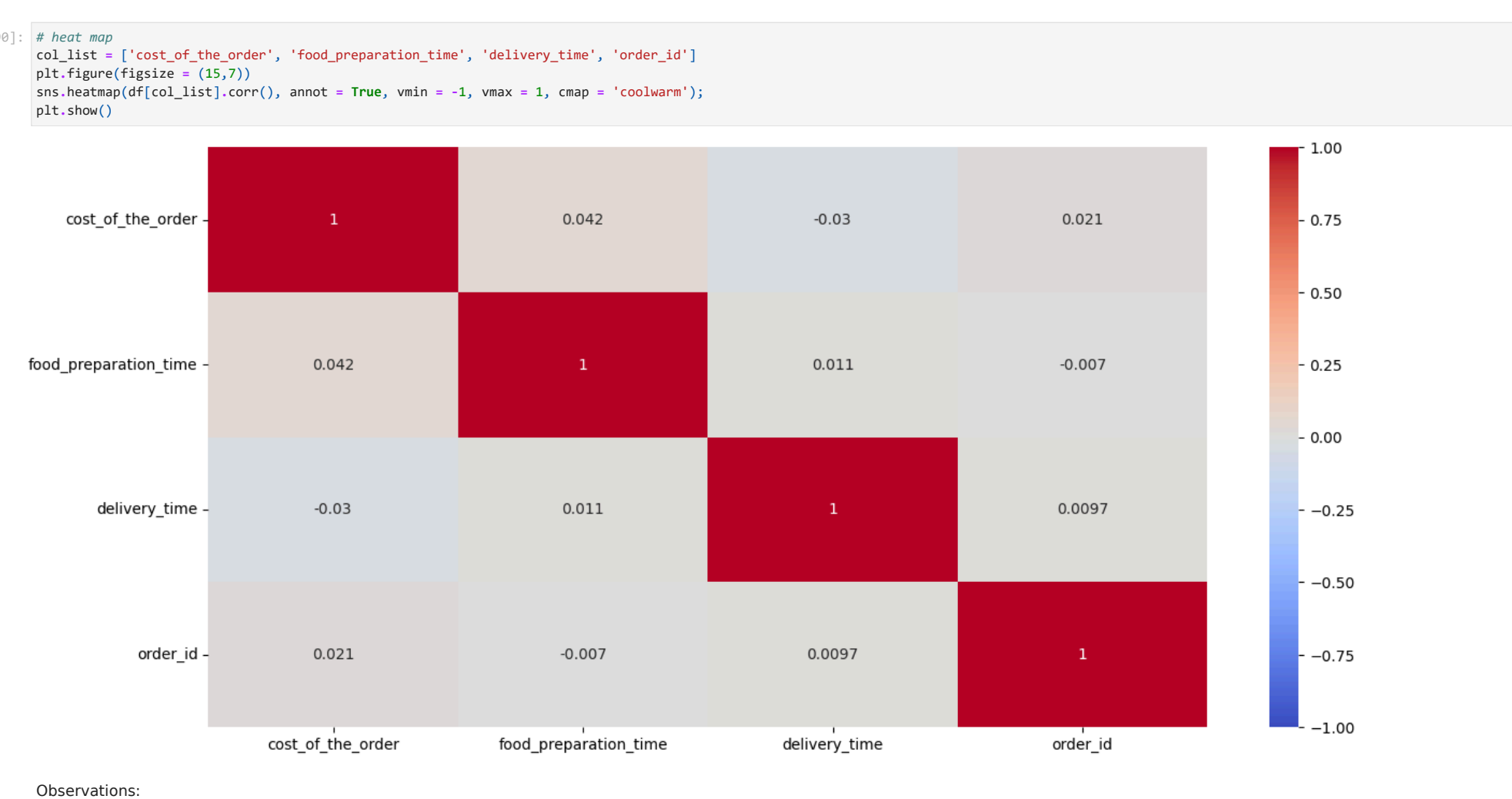
In [89]:
```
#point plot of rating and cost of order
plt.figure(figsize = (15,7))
sns.pointplot(x = 'rating', y = 'cost_of_the_order', data = df);
plt.show()
```



Observations:

- There might be a correlation between the cost of the roder and customer satisfaction

In [90]:
```
# heat map
col_list = ['cost_of_the_order', 'food_preparation_time', 'delivery_time', 'order_id']
plt.figure(figsize = (15,7))
sns.heatmap(df[col_list].corr(), annot = True, vmin = -1, vmax = 1, cmap = 'coolwarm');
plt.show()
```



Observations:

- There is no correlation between cost of order, food preparation time, delivery time, and order ID

Question 13: The company wants to provide a promotional offer in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4. Find the restaurants fulfilling the criteria to get the promotional offer. [3 marks]

In [91]:
```
#Filter out restuarant names
df_rating = df[df['rating'] != 'Not given'].copy()
#Convert rating from object to an integer
df_rating['rating'] = df_rating['rating'].astype('int')
#Create a new data set with restaurant names and ratings
df_rating_count = df_rating.groupby(['restaurant_name'])['rating'].count().sort_values(ascending = False).reset_index()
df_rating_count.head(5)
```

Out[91]:

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64
4	RedFarm Broadway	41

In [92]:
```
# Get restaurant names that have ratings over 50
rest_names_more_than_50 = df_rating_count[df_rating_count['rating'] > 50]
rest_names_more_than_50
```

Out[92]:

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64

In [93]:
```
#Fliter the data
rest_rating_more_than_50 = df_rating[df_rating['restaurant_name'].isin(rest_names_more_than_50['restaurant_name'])]
rest_rating_more_than_50
```

Out[93]:

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time	order_count
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15	1
12	1476966	129969	Blue Ribbon Fried Chicken	American	24.30	Weekend	5	23	17	1
15	1477414	66222	Shake Shack	American	16.20	Weekend	5	33	25	1
19	1477354	67487	Blue Ribbon Sushi	Japanese	16.20	Weekend	4	35	26	1
22	1478287	150599	Shake Shack	American	29.10	Weekday	5	21	30	1
...	...	...	...	...	...	...	...	...	...	...
1880	1477466	222734	Shake Shack	American	13.97	Weekend	5	35	27	1
1884	1477437	304993	Shake Shack	American	31.43	Weekend	3	31	24	1
1885	1477550	97324	Shake Shack	American	29.05	Weekday	4	27	29	1
1886	1478277	48282	The Meatball Shop	American	14.50	Weekend	5	34	25	1
1888	1477353	106324	The Meatball Shop	Italian	16.20	Weekend	5	21	26	1

354 rows × 10 columns

In [94]:
```
# Group the restaurant names with the rating and find the mean of the rating
rest_rating_more_than_50.groupby(['restaurant_name'])['rating'].mean().sort_values(ascending = False).reset_index().dropna()
```

Out[94]:

	restaurant_name	rating
0	The Meatball Shop	4.511905
1	Blue Ribbon Fried Chicken	4.328125
2	Shake Shack	4.278195
3	Blue Ribbon Sushi	4.219178

Observations:

- The four restaurants that will receive the promotion are The Meatball Shop, Blue Ribbon Fried Chicken, Shake Shack, and Blue Ribbon Sushi

Question 14: The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars. Find the net revenue generated by the company across all orders. [3 marks]

In [95]:
```
#create function to calculate net revenue
def net_revenue(x):
  if x > 20:
    return x * 0.25
  elif x > 5:
    return x * 0.15
  else:
    return 0
df['revenue'] = df['cost_of_the_order'].apply(net_revenue)
df.head(5)
```

Out[95]:

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time	order_count	revenue
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20	1	7.6875
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23	1	1.8120
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28	1	1.8345
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15	1	7.3000
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24	1	1.7385

In [96]:
```
# Add up all the revenues
total_net_revenue = df['revenue'].sum()
print('The net revenue generated by the company across all orders is: $', total_net_revenue)
```

```
The net revenue generated by the company across all orders is: $ 6166.303
```

Observations:

- The total net revenue generated is $6166.30

Question 15: The company wants to analyze the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed? (The food has to be prepared and then delivered.) [2 marks]

In [97]:
```
# Add up food prepartaion time and delivery time
df['total_delivery_time'] = df['food_preparation_time'] + df['delivery_time']
# Find the values that are greater than 60
df_greater_than_60 = df[df['total_delivery_time'] > 60]
# Divide values greater than 60 by all values and mulitply by 100 to get a percentage
delivery_percent = (df_greater_than_60.shape[0] / df.shape[0]) * 100
print('The percentage of orders that take more than 60 minutes to get delivered from the time the order is placed is', round(delivery_percent, 2), '%')
```

```
The percentage of orders that take more than 60 minutes to get delivered from the time the order is placed is 10.54 %
```

Observations:

- 10.54% of the total orders take longer than 60 minutes to get delivered
- This could be due to customers ordering foods that take longer to prepare, traffic, or greater distance to cover for the delivery driver
- This also means that around 90% of the orders take less than hour to prepare and deliver

Question 16: The company wants to analyze the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends? [2 marks]

In [98]:
```
# Get the mean delivery time for weekend and weekdays
weeknd_times = df[df['day_of_the_week'] == 'Weekend']['delivery_time'].mean()
weekdays_times = df[df['day_of_the_week'] == 'Weekday']['delivery_time'].mean()
# Print answer
print('The mean delivery time during weekdays is:', round(weekdays_times, 2), 'minutes')
print('The mean delivery time during weekends is:', round(weeknd_times, 2), 'minutes')
```

```
The mean delivery time during weekdays is: 28.34 minutes
The mean delivery time during weekends is: 22.47 minutes
```

Observations:

- The mean delivery time during weekdays is 28.34 minutes
- The mean delivery during weeends is 22.47 minutes
- Delivery time is shorter on the weekends depite having more orders.This could be due to less traffic on the weekends

Conclusion and Recommendations

Question 17: What are your conclusions from the analysis? What recommendations would you like to share to help improve the business? (You can use cuisine type and feedback ratings to drive your business recommendations.) [6 marks]

Conclusions:

- Shake Shack has the highest number of orders at 133, making it the most popular restaurant
- The Meatball Shop has the highest rating of any other restaurant, at 4.5, mkaing it the most liked restaurant
- The Top 5 cuisine types are American, Japanese, Italian, Chinese and Mexican, with these 5 cusinies taking up 87% of the total orders
- 736 orders were not given ratings, which could affect the overall ratings of the restaurants
- Weekends appear to be busier than weekdays, with an increase in total orders for the weekend
- About 10.54% of orders take longer than 1 hour

Recommendations:

- Foodhub should focus on promoting restuarants like Shake Shack and The Meatball Shop, since they are the most popular and highest rated restaurants respectively. They can do so by implmenting ads into the app and offer coupons, or discounts, for these restaurants to bring in more customers
- Foodhub should look to recurit more restaurants that serve American, Japanese, Italian, Chinese, and Mexican cuisines, since these foods take up the vast majority of the total orders.
- Foodhub should implement a rating feature in which customers can rate their order, the restaurant, and overall satisfaction after each order is completed. This can allow for customers to leave more reviews and will thus drastically decrease the amount of 'not given' ratings per order
- Weekdays are less profitable than weekends. So, Foodhub should provide promotions for weekdays to give customers an incentive to order during the week. This will increase the amount of total orders for weekdays.

- The amount of ordesr increase during the weekend. For this, more delivery drivers should be scheduled to work weekends to keep up with the high demand of orders.
- 10.54% of orders take over an hour to deliver, which could be an issue for customer satisfaction. To prevent customers from becoming dissatisfied, Foodhub should offer free delivery for order that take more than 1 hour, or work to get delivery time down to meet customer expectations.
